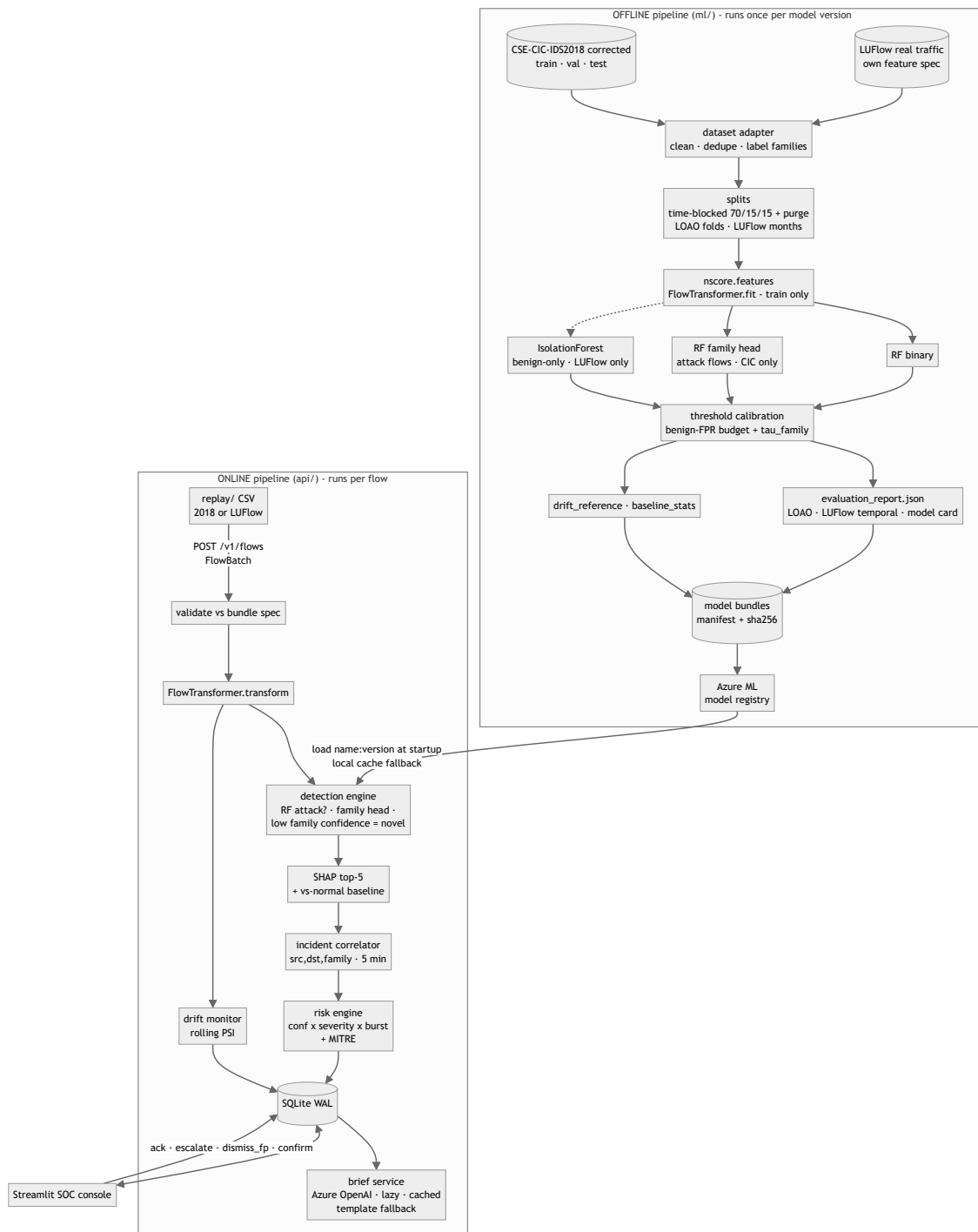


NetSentinel Final Architecture

One line: NetSentinel detects attacks with a Random Forest and names them with a second forest; when it detects an attack but can't say which family, it calls it **unfamiliar** (a novel variant). The output is grouped into explained, **risk-scored incidents** for a SOC analyst. *Measured, not assumed: a benign-only anomaly detector did not work on these flows, while the supervised forest generalised to attacks it had never seen (docs/experiments.md).* It's trained and tested on CSE-CIC-IDS2018 and shown working on real traffic (LUFlow). Nothing is ever auto-blocked.

1. System overview



Design principles

1. **Separate offline and online pipelines** (kept from Doc B). Serving never imports training-only dependencies.
2. **One shared core (nscore/)** for anything that must behave the same in training and serving: feature transform, fusion, risk engine, drift math, bundle loading. This prevents train/serve skew by construction.
3. **Spec-driven, so dataset-agnostic.** Every bundle carries its own `feature_spec.json`. The same code trains and serves a CIC-schema model (2018) and a LUFlow-schema model, with no dataset-specific branches downstream of the adapters.

4. **Contract first.** `nscore/contracts/` (schemas, policy, fixtures) is the only coupling between components, so five people can build in parallel.
5. **Honest by default.** Thresholds come from an FPR budget, every claim has a matching metric, and the limitations are written down.
6. **Demo-proof.** Every external dependency (Azure ML, Azure OpenAI, Wi-Fi) has a local fallback.

2. Repository layout

```

netsentinel/
├─ nscore/
│   └─ contracts/      shared core library (pip install -e .)
│   └─ features/       schemas.py · policy.py (risk engine) · feature_spec*.json · fixtures/ ← the contra
│   └─ detection/      FlowTransformer (M1)
│   └─ drift/          fusion.py (M2)
│   └─ bundle/         PSI + reference builder (M1)
│                       manifest, packager, loader w/ Azure ML + cache (M2)
├─ ml/                offline pipeline: data/ (adapters/) train/ evaluate/ explain/ registry/ (M1, M2)
├─ api/               FastAPI service: app/ (routers, services, repo) tests/ (M3, M5 brief)
├─ dashboard/         Streamlit SOC console: app.py, pages/ (M4)
├─ replay/            replay engine + scenarios/*.yaml (M5)
├─ infra/             docker-compose, Azure provisioning scripts (M3)
├─ notebooks/         EDA + experiments (never imported by code)
├─ data/              gitignored: raw/ processed/ splits/ replay/
├─ artifacts/          gitignored: bundles, cache
├─ tests/             cross-cutting tests (contracts, train/serve parity)
└─ docs/              this folder + model_card.md + threat_model.md + runbooks

```

3. Offline pipeline

3.0 Datasets: two datasets, two jobs

The PS names datasets "only as examples"; the rule is free and public.

Dataset	Job	Why this one	Feature schema
CSE-CIC-IDS2018, corrected (Liu, Engelen et al., IEEE CNS 2022)	Train, validate, test (P1, P2)	A large AWS network (420 machines, 30 servers, 50 attacker machines), 10 capture days, 7 attack scenarios with many tools (Patator, Hulk, GoldenEye, Slowloris, LOIC-HTTP/UDP, HOIC , DVWA web attacks, infiltration, Ares botnet). Labels were manually audited and the extractor bugs fixed.	CIC (fixed CICFlowMeter, ~80 features)
LUFlow (Lancaster University honeypots, labelled via threat intelligence)	Real-world showcase (P3): its own bundle, served live in the demo	Real internet attack traffic on a real university network, collected continuously since 2020, so it has real drift. Its outlier label ("abnormal but unexplained") is exactly what the novelty detector targets.	LUFlow (16 CSV fields → 9 features): <code>feature_spec.lufLOW.json</code>

Downloads: distrinet-research.be/CNS2022 (corrected 2018 + fixed CICFlowMeter) · github.com/ruzzzzz/LUFlow (or Kaggle).

2018 attack schedule (from the CIC dataset page), which drives the split design. Each attack runs for about 1 hour inside a normal working day:

Day	Attacks
Wed 14-02	FTP-BruteForce, SSH-BruteForce
Thu 15-02	DoS-GoldenEye, DoS-Slowloris

Day	Attacks
Fri 16-02	DoS-SlowHTTPTest, DoS-Hulk
Tue 20-02	DDoS-LOIC-HTTP, DDoS-LOIC-UDP
Wed 21-02	DDoS-LOIC-UDP, DDoS-HOIC
Thu 22-02, Fri 23-02	Web: Brute Force, XSS, SQL Injection
Wed 28-02, Thu 01-03	Infiltration (malicious download → internal Nmap scan)
Fri 02-03	Botnet (Ares)

Size (measured): 63.2M raw flows (45.2M after removing exact duplicates), 36 GB of CSV inside a 10.4 GB zip. M1 never unzips: it streams each CSV in ~400 MB in-memory chunks (the streaming reader on a zip member was 40x slower), writes float32 parquet (8 GB), and builds a fixed-seed working sample of 3.5M train / 1.9M validation / 2.0M test flows, which trains comfortably on a 16 GB laptop. Full counts, dataset quirks and every drop decision: docs/data_profile.md . **LUFlow:** 63.0M raw flows in the 72-day subset (44.9M clean). About 10% of its time_start values are corrupted by the source and are repaired in the adapter; 9 features survive the spec.

Check on first download: the corrected files must contain Src IP , Dst IP and Timestamp (the incident correlator and the time-blocked split need them). The original 2018 CSVs lack IPs on most days. That's one more reason to use only the corrected release.

Dataset adapters (ml/data/adapters/{cic2018,luflow}.py) turn each source into one canonical frame: metadata columns + features (named per the matching feature spec) + family + period (day or month). Everything downstream reads only that frame.

3.1 Data → families

- Drop from features, keep as metadata: Flow ID , Src IP , Dst IP , Src Port , Timestamp . Dst Port is an open experiment: run it both ways and document the result (B12).
- Inf/NaN: count them per column, then drop or clip them, and record the counts in data_profile.md . Remove exact duplicates.
- **2018 label mapping** → AttackFamily (counts: data/README.md): SSH-BruteForce → BruteForce (FTP-Patator is 100% Attempted , so it is benign) · DoS GoldenEye/Slowloris/Hulk → DoS (SlowHTTPTest is absent from the corrected release) · DDoS LOIC-HTTP/LOIC-UDP/HOIC → DDoS · Web Brute Force/XSS/SQL Injection → WebAttack · Infiltration (all stages) → Infiltration · Ares → Botnet . "Attempted" labels in the corrected set → BENIGN , as the dataset authors recommend. Never treat them as their own class.
- **Low-support rule:** any family with < 1,000 clean flows is flagged low-support. Measured: WebAttack has only 283 flows (every other family has ≥ 89k). It stays a named family (keeps its own severity 0.6 and MITRE T1190), uses class_weight , is reported with a caveat, and is **excluded from LOAO**. Merge it into Rare only if M2 finds it unlearnable.
- **LUFlow mapping:** benign → BENIGN , malicious → attack (binary only; the family is Malicious), outlier → left out of supervised training , kept as a separate label to measure novelty capture.
- Correlation pruning ($|p| > 0.95$, keep the more interpretable feature) on the 2018 train split only → nscore/contracts/feature_spec.json .

3.2 Evaluation protocol

Protocol	How	Answers	Used for
P1 time-blocked + purge (2018)	For each (day, label) group, sort by timestamp: first 70% → train, next 15% → val, last 15% → test. Drop flows within 60 s of each block boundary (purge) so a single attack session can't sit on both sides. Indices are saved to <code>data/splits/</code> so every run uses the exact same split.	How well do we detect <i>known</i> families?	Main per-class metrics, threshold tuning (val), model card
P2 LOAO (2018)	For each family F in {DoS, DDoS, BruteForce, Infiltration, Botnet} (WebAttack is too small for a held-out test): remove F completely from training of BOTH heads, retrain on a fixed-size sample, then measure recall on F's test flows at benign-FPR budgets 0.01-0.5% (3 seeds, mean and range), and how many detections the family head labels novel .	Can we catch a family we've never seen?	The headline chart: held-out recall per family vs the same model trained WITH it (<code>seen ceiling</code>). Result: floods, DoS and botnet generalise; SSH brute force and internal scans do not
P2b tool holdout (2018)	Retrain without ONE tool whose siblings stay (DDoS-HOIC, LOIC-HTTP, Hulk, GoldenEye, Slowloris). HOIC is excluded from hyper-parameter tuning so it is an untouched check.	Does the model learn behaviour , not tools? Does it flag an unfamiliar <i>variant</i> of a known family?	<code>kind=tool</code> rows in the report + a Q&A answer
P3 real-world temporal (LUFlow)	Same pipeline with <code>feature_spec.luflow.json</code> : train RF + IForest on the earliest months, test month by month afterwards (recall, FPR, PSI over time). Also measure the share of <code>outlier</code> flows flagged by the IForest vs the RF (the only place the benign-only detector is evaluated against real unexplained traffic).	Does it hold on real traffic, and how fast does it decay?	Real-world showcase, measured drift curve
P3b recalibration (LUFlow)	Refit only the IF + thresholds on a label-free recent benign window from a later month, then re-test that month.	Can we recover from drift without new labels?	Drift → recalibrate → recover story, demo act 5

Metrics reported per class: precision, recall, F1, **FPR**, support, one-vs-rest ROC-AUC. Overall: macro-F1, binary ROC-AUC and PR-AUC, benign FPR at the operating point. Confusion matrix saved as PNG and JSON. Probability calibration checked with a reliability plot. P2b and P3/P3b go into `EvaluationReport.external` .

Why not a day-based split: in 2018, as in 2017, each attack lives on one or two specific days. Splitting by day would leave whole families out of training. Leaving a family out on purpose is what LOAO does, deliberately and measurably.

3.3 Models

Model	Train on	Role	Notes
<code>rf_bin</code> <code>ary</code>	all train flows	p(attack)	100 trees, depth 16, 100 samples per leaf, 20% of features per split, no class weights. Tuned on held-out-tool recall (in-distribution accuracy is saturated); class weights vs SMOTE made no measurable difference. Tiny: ~5 MB.
<code>rf_mult</code> <code>iclass</code>	attack flows only	which family + probabilities per family (feed expected severity) and its confidence (max probability), which is the novelty signal	CIC bundles only. LUFlow has no family labels, so its bundle has <code>family_head = False</code> .

Model	Train on	Role	Notes
iforest	benign train flows only	anomaly percentile of benign val scores	Optional, LUFlow bundles only. On CIC flows it scored ROC-AUC 0.70-0.87 and ~0 recall at a usable false-alarm rate (docs/experiments.md §4), so CIC bundles do not ship one.

Thresholds (M2-07): `tau_binary` = the lowest threshold whose benign FPR on validation is $\leq$ the **operating budget (0.1%)**; `tau_family` = the family-confidence level below which only 2% of *familiar* validation attacks fall (so about 2% of familiar attacks get a false "novel" label, measured on test). The full trade-off (budgets 0.01%-1%) ships in the bundle as `operating_curve.json`. Why 0.1% and not the 1% first planned: at this network's scale (~330k benign flows per hour) 1% is thousands of false alarms an hour and cuts precision at natural prevalence from 99.9% to 87%. Even 0.1% is hundreds per hour before the correlator groups them (§4.3). **All precision and counts are weighted to the real class balance** (the val/test samples keep every attack but only a quarter of benign flows).

3.4 Explainability

- `shap.TreeExplainer` on `rf_binary`: it explains *why the flow was flagged as an attack*, which holds for known and novel verdicts alike. Exact TreeSHAP costs ~50-110 ms per flow, so the API explains a few representative flows per incident (`fast` mode = first 25 trees) rather than every flow of an alert storm.
- Each alert gets the top 5 `{feature, raw value, shap_value, benign median}`. The benign median comes from `baseline_stats.json` and lets the UI say "this flow's inter-arrival time is 4,000× shorter than normal".
- Global importance is precomputed for the Model page.
- SHAP never changes the risk score.** One number decides the order, one explanation builds trust.
- Latency budget: detection is milliseconds; SHAP is the slow part (see above).

3.5 Model bundle (the offline → online contract)

bundle_vN/	
manifest.json	BundleManifest: version, dataset, split, feature_spec sha256, file sha256s, metrics
feature_spec.json	ordered features, clip/log rules (CIC or LUFlow schema)
transformer.joblib	fitted FlowTransformer
rf_binary.joblib	
rf_multiclass.joblib	(absent when family_head = False)
iforest.joblib	OPTIONAL (LUFlow bundles)
iforest_benign_val_scores.npy	OPTIONAL, for score → percentile
thresholds.json	<code>{"tau_binary", "tau_family", "operating_fpr"}</code> (+ "tau_anomaly" with an IForest)
operating_curve.json	benign FPR / recall / precision per false-alarm budget, per-family recall
global_importance.json	mean SHAP per feature (Model page)
label_map.json	class index → AttackFamily
drift_reference.json	quantile bins + proportions, top-15 features + reference attack rate
baseline_stats.json	benign median / p95 per feature
evaluation_report.json	EvaluationReport schema
confusion_matrix.png, loao.png, reliability.png	

Bundles in the registry (each one versioned and tagged):

Bundle	Trained on	Used for
netsentinel-bundle:N	2018, all families	Main demo and evaluation
netsentinel-demo-holdout-botnet:N	2018 without Botnet	Demo act 4: "this model has never seen a botnet"
netsentinel-luflow:N	LUFlow earliest months	Demo act 5: real traffic

Bundle	Trained on	Used for
netsentinel-luflow-recal:N	same RF, IF + thresholds refit on a recent benign window	Demo act 5: drift recovery

The manifest's `feature_spec_sha256` stops the API from scoring CIC flows with a LUFlow bundle, and vice versa.

3.6 Registry

Azure ML workspace. `MLClient.models.create_or_update` with tags `{dataset, split, contract_version, feature_schema, macro_f1, benign_fpr, held_out}`. The API resolves `MODEL_REF=azureml:netsentinel-bundle@latest`, downloads it to `artifacts/cache/`, and verifies the sha256s. If Azure is unreachable, it falls back to the cached copy. `local:` refs are supported for development.

4. Online pipeline (FastAPI)

4.1 Endpoints (/v1 , all bodies are contract models)

Method	Path	Body / query	Returns	Notes
GET	/health		{status, model_loaded, model_version}	liveness + readiness
GET	/v1/model		ModelInfo	includes feature_schema and family_head , so the UI adapts
GET	/v1/model/evaluation		EvaluationReport	feeds the Model page
POST	/v1/flows	FlowBatch (≤ 500) · header X-API-Key	ScoreBatchResponse	validates against the loaded bundle's feature spec → transform → fuse → SHAP (non-benign only) → correlate → risk → persist
GET	/v1/incidents	status, verdict, family, level, since, limit, offset, sort=risk\ last_seen	IncidentPage	
GET	/v1/incidents/{id}		IncidentDetail	
POST	/v1/incidents/{id}/actions	AnalystActionIn · header X-Analyst	AnalystActionRecord	updates status, writes audit log
GET	/v1/incidents/{id}/brief	refresh=false	Brief	lazy, cached, 8 s timeout → template fallback
GET	/v1/drift		DriftReport	
GET	/v1/metrics		LiveMetrics	throughput, p50/p95 latency, confirmed precision, MTTA
POST	/v1/admin/reload-model	{model_ref} · admin key	ModelInfo	switches the demo between bundles (2018 → holdout → LUFlow → LUFlow-recal) and starts a fresh drift window

Mock mode (NS MOCK=1): every GET serves nscore/contracts/fixtures/*.json . This is ready on day 1 so the dashboard isn't waiting on the backend.

4.2 Scoring path for one flow

```
FlowRecord → transformer.transform_records → x (rejects flows missing spec features: 422)
p = rf_binary.predict_proba(x)[1]
if p >= tau_binary and family_head:
    probs = rf_multiclass.predict_proba(x); closest = argmax(probs); fconf = max(probs)
v = fuse(p, a, tau_binary, tau_anomaly, fconf, tau_family) # a = IForest percentile (LUFlow only, else 0)
if v == KNOWN_ATTACK: family = closest if family_head else Malicious; conf = p
if v == NOVEL_ANOMALY: family = Unknown (closest_family = closest, family_confidence = fconf); conf = p
if v != BENIGN:
    sev = policy.expected_severity(family, probs)
    top5 = shap(explainer_for(v), x) joined with baseline_stats
    incident = correlator.upsert(meta, v, family, conf, sev, top5)
    incident.risk = policy.risk_score(v, incident.max_confidence, incident.severity, incident.flow_count)
drift_monitor.observe(x, v) # every flow, benign included
```

4.3 Incident correlator

- Key: (src_ip, dst_ip, family) . Novel anomalies use (src_ip, dst_ip, "Unknown") .
- Window: an open incident absorbs a matching flow if flow.observed_at - incident.last_seen ≤ 5 min and the incident's status isn't resolved or dismissed_fp .
- On upsert: flow_count += 1 , max_confidence = max(...) , severity = running mean of the flows' expected severity, last_seen , a running mean of |SHAP| per feature for the top 5, and a recomputed risk.
- The effect: replaying a 50k-flow DDoS-HOIC attack produces about one incident per source/destination pair. Report the flows-to-incidents ratio. It's a strong slide.

4.4 Risk engine (nscore/contracts/policy.py)

This merges the team's **Risk Scoring Pipeline** doc (docs/reference/NetSentinel_Risk_Scoring_Pipeline.pdf) with the incident and novelty design. **Confidence measures certainty, severity measures consequence**: a model can be very sure about something harmless.

```
severity = Σ_f P(f | attack) × W[f] expected severity over the family head's probabilities
          binary-only bundle (LUFlow) → W[Malicious] · novel anomaly → W[Unknown]
confidence = p_attack (known attack) | novel_confidence(anomaly percentile) (novel anomaly)
burst = 1 + 0.15 × min(1, log10(flow_count) / 3) 1 flow → 1.00 · ≥1000 flows → 1.15
risk = min(100, round(100 × (confidence × severity × burst + novelty_bonus))) novelty_bonus = 0.10
level = HIGH ≥ 70 · MEDIUM 40-69 · LOW < 40
```

Family	W	Why (M5 writes the full reasoning in docs/threat_model.md)
Infiltration	1.00	Deepest stage: the attacker is already inside
Botnet	0.90	A host is likely already compromised
DoS / DDoS	0.80	Service disruption, high visibility
Rare	0.80	Merged rare families: treat as serious until triaged
Unknown (novel)	0.75 + 0.10 bonus	Unknown intent, so nothing is assumed safe
BruteForce	0.70	Active attempt to gain access
Malicious (LUFlow)	0.70	Malicious per threat intelligence, type unknown

Family	W	Why (M5 writes the full reasoning in docs/threat_model.md)
WebAttack	0.60	Could mean an exploited app, but often fails
PortScan	0.30	Reconnaissance only (not a 2018 class; kept for live capture)

What changed vs the team doc, and why

- **Expected severity** instead of the single top-family weight. If the classifier is split 50/50 between WebAttack (0.6) and Infiltration (1.0), severity = 0.8, not a coin flip between the two extremes.
- **The burst factor (the doc's Factor 3) is in the baseline**, applied per incident through the correlator, as a multiplier in [1.00, 1.15]. A single flow gets exactly the doc's formula, so the doc's worked examples still hold (62 MEDIUM, 81 HIGH, 28 LOW, all asserted in tests/test_contracts.py).
- **Novel anomalies get a score**, instead of being dropped at the 0.5 gate.
- **The FPR-budget threshold replaces the fixed 0.5.**

Worked examples (from the fixtures): Infiltration, 1 flow, conf 0.81 → **81 HIGH** · DDoS-HOIC, 4,210 flows, conf 0.99 → **91 HIGH** · Botnet, 37 flows, conf 0.94 → **91 HIGH** · novel anomaly, 112 flows, conf 0.71 → **69 MEDIUM** · BruteForce, 1 flow, conf 0.88 → **62 MEDIUM** · unsure WebAttack (90/10 with Infiltration), conf 0.58 → **37 LOW**.

4.5 Database (SQLite, WAL mode; swap the URL for Azure SQL or Postgres later)

```
CREATE TABLE flows (
  flow_id TEXT PRIMARY KEY, observed_at TEXT, received_at TEXT,
  src_ip TEXT, dst_ip TEXT, src_port INT, dst_port INT, protocol INT,
  features_json TEXT, verdict TEXT, p_attack REAL, anomaly_pct REAL,
  attack_family TEXT, severity REAL, incident_id TEXT REFERENCES incidents(incident_id),
  ground_truth TEXT, model_version TEXT, latency_ms REAL
);
CREATE TABLE incidents (
  incident_id TEXT PRIMARY KEY, status TEXT DEFAULT 'new', verdict TEXT, attack_family TEXT,
  mitre_id TEXT, mitre_name TEXT, risk_score INT, risk_level TEXT, severity REAL, max_confidence REAL,
  flow_count INT, src_ip TEXT, dst_ip TEXT, dst_port INT, first_seen TEXT, last_seen TEXT,
  top_features_json TEXT, brief_json TEXT, model_version TEXT,
  acknowledged_at TEXT, updated_at TEXT
);
CREATE TABLE analyst_actions (
  action_id INTEGER PRIMARY KEY AUTOINCREMENT, incident_id TEXT REFERENCES incidents(incident_id),
  analyst TEXT NOT NULL, action TEXT NOT NULL, note TEXT, at TEXT DEFAULT CURRENT_TIMESTAMP
);
CREATE TABLE drift_snapshots (
  id INTEGER PRIMARY KEY AUTOINCREMENT, computed_at TEXT, model_version TEXT, window_size INT,
  status TEXT, max_psi REAL, report_json TEXT
);
CREATE INDEX ix_inc_open ON incidents(status, risk_score DESC);
CREATE INDEX ix_inc_key ON incidents(src_ip, dst_ip, attack_family, last_seen);
```

Benign flows: keep at most N recent ones (configurable), because drift only needs the rolling window.

4.6 Drift monitor

- Rolling window of the last 2,000 flows **the model calls benign**. Every 500 flows it computes PSI per feature against the **loaded bundle's** drift_reference.json , which is built from **benign training flows only**, plus the predicted attack rate over all flows. It answers "does normal traffic still look like what the model learned?"; attacks are reported as incidents. (A reference mixing in the training attacks made random normal traffic read PSI 1.36 = ALERT; benign-only reads 0.02.) Fewer than 200 benign flows in the window (an attack burst): no new snapshot.

- Status: max PSI < 0.10 is `ok` , < 0.25 is `watch` , anything higher is `alert` . Each check writes a snapshot. Alert status shows a "consider recalibrating" banner linked to `docs/runbook_retrain.md` .
- Within 2018 a short replay slice can still read `watch` / `alert` on timing features (one busy 72-second slice vs a 10-day average: max PSI ~0.30). **The real drift story is LUFlow**: February 2021 through the June-July 2020 bundle reads `alert` (max PSI 0.33) while detection holds (precision 98.6%, recall 99.9%). The recalibrated bundle does not lower PSI and did not improve detection (`docs/experiments.md` §8).

4.7 Brief service (Azure OpenAI)

- Input: structured incident JSON only (family, verdict, confidence band, risk level, flow count, top features with raw values and benign medians, MITRE ID, IPs and port). No raw payloads.
- System prompt (extends Doc B §6): write 3–4 sentences covering what was seen, why it was flagged (only from the given features), and one suggested next step. Don't invent details. **Hedge to match the confidence band**. For `novel_anomaly` , say plainly that no known family matched. For `Malicious` (LUFlow), don't guess a type.
- Generated when first requested, then cached in `incidents.brief_json` . Times out after 8 s and falls back to a deterministic template brief (`source: "template"`). A failed brief never blocks the queue.
- Config comes only from environment variables: `AZURE_OPENAI_ENDPOINT` , `AZURE_OPENAI_API_KEY` , `AZURE_OPENAI_DEPLOYMENT` , `AZURE_OPENAI_API_VERSION` .

4.8 Security and ops of the system itself

- `X-API-Key` on ingest, `X-Analyst` (name + role) on actions, admin key on reload. Secrets live in `.env` , never in git, and CI runs secret scanning.
- Structured JSON logs with a request ID. p50/p95 latency is exposed in `/v1/metrics` .
- Pydantic validation rejects malformed flows with a 422 that names the missing feature.

5. SOC console (Streamlit)

Page	Shows	Key interactions
Live Queue	Incidents sorted by risk score . HIGH red, MEDIUM amber, LOW grey. Chips for <code>Known attack</code> and <code>Novel anomaly</code> . Header counters: open HIGHs, novel anomalies, flows/sec. Auto-refreshes every 3 s and highlights new rows. A badge shows which bundle is loaded (2018 / LUFlow).	filter by status/verdict/family/level, click through to detail
Incident Detail	Risk breakdown (confidence × severity × burst = score) · SHAP bar chart · "vs normal" table (value vs benign median) · MITRE badge · flow metadata · brief panel (loading, LLM or template label) · action timeline	Acknowledge / Escalate / Confirm / Dismiss as FP, with a note
Model & Evaluation	Model version and registry ref · per-class table · confusion matrix · LOAO chart · Real-world panel (LUFlow month-by-month + recalibration, from <code>external</code>) · operating FPR · limitations	—
Drift & Health	PSI per feature against the 0.10/0.25 lines · status banner · throughput · latency	—
Analyst Metrics	Analyst-confirmed precision · FP dismiss rate · MTTA · actions per analyst	—

When `family_head` is false (LUFlow bundle), the UI hides family-specific widgets (MITRE, confusion matrix) instead of showing empty ones. Analyst sign-in is a name plus a role in session state, sent as `X-Analyst` . React stays a stretch goal and uses the same API.

6. Demo storyline (≈ 5 min)

1. **Calm.** 2018 benign replay runs, the queue is empty, drift is `ok`, and the model page shows v N from Azure ML.
2. **Known attack.** A brute-force burst arrives as one incident, mapped to MITRE T1110. Open it and show the **risk breakdown** ("88% sure × severity 0.7 × burst"). Generate the brief, then Escalate.
3. **Alert storm.** Replay DDoS-HOIC: thousands of flows become **one HIGH incident**. "That's the alert-fatigue answer."
4. **The novel attack.** Reload the **holdout bundle** (trained without any botnet) and replay Ares botnet traffic. The forest still flags it, but the family head can't place it, so the incident appears as **Novel anomaly** ("closest known family: X, 55% sure") with a SHAP explanation. Then reload the full bundle: the same traffic becomes a named **Botnet** incident. Show the LOAO chart: "measured across every family, not just this demo".
5. **Real traffic.** Reload the **LUFlow bundle** and replay real honeypot traffic from a **later month**. Real attackers show up as incidents, unexplained `outlier` traffic gets flagged as novel, and **drift jumps to alert** because the internet changed since training. Reload **LUFlow-recal** (refit on a recent benign window, no labels needed) and drift drops back to `ok`. Show the month-by-month chart.
6. **Honesty.** Dismiss a low-confidence incident as a false positive and the analyst-confirmed precision updates live. Walk through the limitations slide.
7. Fallbacks: a recorded video of acts 1–6. Briefs fall back to the template if Azure OpenAI is unreachable. Bundles load from the local cache if Azure ML is unreachable.

7. Tech stack

Layer	Choice	Why
ML	scikit-learn, imbalanced-learn, shap, pandas/pyarrow	It's what the PS names, it's fast on a laptop, and TreeSHAP is exact and quick
Registry	Azure ML (SDK v2) + local cache	Real use of Microsoft tooling, with a demo fallback
API	FastAPI, Uvicorn, Pydantic v2, SQLAlchemy Core / sqlite3	Contract-typed and quick to build
LLM	Azure OpenAI (gpt-4o-mini class deployment)	Cheap, quick, and a Microsoft service
UI	Streamlit + Plotly	Fast to build. React is a stretch goal
Infra	Docker Compose (api + dashboard). Optional deploy to Azure Container Apps	One command runs the demo
CI	GitHub Actions: ruff, pytest, gitleaks	Required checks on PRs

8. Requirements traceability

PS requirement	Where it's met	Evidence on demo day
Normal vs attack + attack types	<code>rf_binary</code> , <code>rf_multiclass</code>	Family label on every 2018 incident
Surfaces novel attacks	binary RF generalisation + family-head confidence (<code>fusion.fuse</code>)	Demo act 4 + leave-one-out tables (floods/DoS yes; botnet and SSH brute force only at 0.1-0.5% budgets; Nmap scans and LOIC-HTTP no) + LUFlow outlier capture (act 5)
Precision / recall / FPR / AUC	<code>ml/evaluate</code> → <code>EvaluationReport</code>	Model page, model card

PS requirement	Where it's met	Evidence on demo day
Class imbalance	class weights, SMOTE comparison, low-support rule	Model card section + before/after table
Discuss model drift	<code>nscore/drift</code> , <code>/v1/drift</code> , drift page, P3/P3b, model card §Drift	Measured decay on real traffic (LUFlow month by month), live drift alert and recovery in act 5
Alert SOC, no auto-block	incidents + risk engine + actions + audit log	Demo acts 2–6
Honest evaluation	purged time-blocked split, FPR-budget thresholds, corrected dataset, limitations	Model card, Q&A sheet
Works beyond lab data	LUFlow bundle	Act 5

9. Key decisions (short ADRs)

#	Decision	Alternatives considered	Why
ADR-1 (revised after measurement)	Binary RF detects; family-head confidence marks unfamiliar attacks. IsolationForest optional, LUFlow only	Original: RF + benign-only IsolationForest fusion; autoencoder	We assumed a supervised RF cannot detect unseen attacks and built a benign-only IForest for that. Measured on CIC-IDS2018: the IForest has ROC-AUC 0.70-0.87 and ~0 recall at a usable false-alarm rate; the supervised RF detects held-out families/tools (floods, DoS, botnet) at strict budgets and the family head's confidence separates unfamiliar from familiar attacks with ROC-AUC 0.997-1.000. Evidence: <code>docs/experiments.md</code> §4-5. Known gaps (test, 3 seeds): internal Nmap scans and the LOIC-HTTP tool are not detected when unseen; SSH brute force (97% at 0.1% budget, 29% at 0.05%) and botnet (66% at 0.1%, 99% at 0.5%) only at looser budgets.
ADR-2	Purged time-blocked per-class split + LOAO	day-based split; random split	A day-based split removes whole families from training. A random split leaks sessions.
ADR-3	Incidents, not per-flow alerts	per-flow alerts	Per-flow alerts make alert fatigue worse, which contradicts the pitch.
ADR-4	Contract-first monorepo with shared <code>nscore</code>	separate repos; no contracts	Lets 5 people work in parallel and avoids train/serve skew.
ADR-5	Streamlit first	React first	Hackathon speed. Both call the same API.
ADR-6	SQLite (WAL)	Postgres	Zero setup. Moving later is just a URL change.
ADR-7	Rebuild SHAP explainers at load	pickle explainers	Pickled explainers break across shap versions. Rebuilding costs milliseconds.
ADR-8	Corrected CSE-CIC-IDS2018 for train/val/test; LUFlow as a live real-traffic showcase with its own bundle	CIC-IDS2017; 2017 + 2018 cross-network; NetFlow-v3 family; UNSW-NB15	2018 is the larger, more varied labelled network with more attack tools and rare-class examples. The labels are audited. LUFlow adds real traffic and real drift , which 2018's scripted benign traffic lacks. One labelled dataset keeps M1's workload realistic. Trade-off: no same-schema cross-network test (accepted).

#	Decision	Alternatives considered	Why
ADR-9	Risk = confidence × expected severity × burst (+ novelty bonus), 3 levels	raw confidence; per-flow scores; P1–P4 bands	Adopts the team's risk doc (confidence ≠ consequence, 3 levels, defensible weights) and adds expected severity, incident-level burst and novelty scoring.
ADR-10	Operating budget 0.1% benign FPR; all precision weighted to natural prevalence	1% budget (first plan); raw-sample precision	1% is thousands of false alarms per hour and 87% precision at real prevalence; the sample-based precision (96%) is inflated by down-sampled benign. 0.1% keeps recall high on floods while staying workable behind the correlator.

Source: docs/03_architecture.md · github.com/Saroy-ctrl/netsentinel